

# Physics — Qobketganlar

Practice Questions with Answers and Explanations

Question: 43

Language: English

## How to use this book

This collection contains all 43 questions of the test “Physics — Qobketganlar”. Every question shows all answer choices, the correct answer highlighted in green, and a detailed explanation below it. Try to answer on your own first, then check yourself. The answer key on the last page allows quick self-testing.

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1. What is the law  $ma = F$ ?

- A. a system of three differential equations solved which (using also the boundary conditions) it is possible to determine the trajectory of a point-like body of mass  $m$
- B. a link between the acceleration and the mass of a generic body
- C. a link between acceleration and the force to which a generic body is subjected ✓
- D. a link between mass and the force to which a generic body is subjected

### EXPLANATION

C) Newton's second law defines force as mass times acceleration ( $F = ma$ ), linking force and acceleration.

2. The intensity of the Earth's gravitational field at 10 Km high

- A. is slightly less than the intensity on the earth's surface ✓
- B. is strictly equal to the intensity on the earth's surface
- C. is not zero, but very small compared to the intensity on the earth's surface
- D. is zero, as demonstrated by the weightlessness of the astronauts

### EXPLANATION

A) Gravitational field strength decreases with distance ( $g = GM/r^2$ ).

3. Two bullets of different mass are fired at the same speed with the same angle and from the same point. Neglecting air resistance and assuming the ground surface is perfectly flat, which of the two takes the longest time to reach the ground?

- A. (a) the one that has less mass
- B. (b) the one with the greatest mass
- C. (c) they will both take the same time ✓
- D. (d) the data provided is insufficient to answer

**EXPLANATION**

C) Time of flight depends only on vertical motion ( $t = 2v_0 \sin \theta / g$ ).

4. We imagine to replace the core of our planet with a material of such high density as to double the mass of the earth, without altering its other quantities such as volume, shape, etc. What would happen to atmospheric pressure?

- A. (a) it would also double ✓
- B. (b) it would remain unchanged
- C. (c) it would be halved
- D. (d) it would become four times larger

**EXPLANATION**

A) Atmospheric pressure depends on Earth's mass ( $P = \rho gh$ , but  $g \propto M$ ).

5. Two balls of equal mass move with equal speeds and in the opposite direction along a straight line. After the impact each sphere reverses the direction of motion. Is it an elastic collision?

- A. (a) Yes ✓
- B. (b) The data of the problem are insufficient for a response
- C. (c) No
- D. (d) The data of the problem are incompatible with the laws of dynamics

**EXPLANATION**

A) Elastic collisions conserve kinetic energy; reversing velocities implies equal speeds before/after.

6. Angular speed 1. does not depend on the radius  $r$  at which it is calculated. 2. is always tangent to the trajectory. 3. is always constant. 4. constantly changes direction.

- A. does not depend on the radius  $r$  at which it is calculated. ✓
- B. is always tangent to the trajectory.
- C. is always constant.
- D. constantly changes direction.

**EXPLANATION**

A) Angular speed ( $\omega$ ) is the same for all points on a rigid body ( $\omega = v/r$ ).

7. The moment of inertia of a rigid body with respect to an axis passing through the center of gravity

- A. (a) it is null only when the object has zero rotation speed
- B. (b) is null because the axis passes through the center of gravity
- C. (c) it is always the same whatever the axis
- D. (d) can assume different values depending on the axis ✓

**EXPLANATION**

D) Moment of inertia ( $I = \sum mr^2$ ) depends on the axis of rotation.

**8.** For a rigid body, initially at rest, to be in static equilibrium

**A.** (a) it is sufficient that the resultant of the external forces is zero

**B.** (b) it is sufficient that the resulting moment of external forces with respect to the center of mass is zero

**C.** (c) it is necessary that the resultant of external forces is zero ✓

**D.** (d) it is sufficient that all external forces are conservative

**EXPLANATION**

C) Static equilibrium requires both net force and net torque to be zero.

**9.** A small object dropped on the earth from a great distance (for example of the order of the earth-moon distance)

**A.** (a) it cannot fall to earth but will become a satellite of it

**B.** (b) describes a uniformly accelerated motion like any other body that falls on the earth from much shorter distances (of the order of a hundred meters)

**C.** (c) describes a uniform rectilinear motion

**D.** (d) does not describe a uniformly accelerated motion given the great distance from the earth ✓

**EXPLANATION**

D) At large distances, gravitational acceleration isn't constant ( $F = GMm/r^2$ ).

**10.** In the parabolic motion of a projectile, in absence of friction,

**A.** (a) the range is maximum for an initial angle equal to  $\pi/2$  of the speed vector with respect to the horizontal axis

**B.** (b) the range is 2 times the abscissa of the point of maximum height reached by the object ✓

**C.** (c) the distance reached by the object depends on the cube of the initial speed value of the object

**D.** (d) the maximum height reached by the object is maximum for an initial angle equal to  $\pi/4$

**EXPLANATION**

B) Range ( $R$ ) =  $(v_0^2 \sin 2\theta)/g = 2 \times (v_0^2 \sin^2 \theta)/(2g) = 2 \times x_{\text{max\_height}}$ .

**11.** A point-like particle, moving at speed  $v$  is subject to a force  $F = b \times v$  where  $b$  is a constant vector. The work done by the force  $F$

**A.** is always zero ✓

**B.** is always different from zero

**C.** is zero just when the point particle moves with circular motion

**D.** depends from the value of  $b$

**EXPLANATION**

A) Work ( $W$ ) =  $F \cdot d = (b \times v) \cdot v = 0$  (cross product is perpendicular to  $v$ ).

- 12.** Shoes with stiletto heels leave a mark on the wooden floors. Why?
- A. (a) because women tend to put their body weight on walking especially on heels
  - B. (b) because the edges of the heel are not sufficiently rounded
  - C. (c) because the material the heels are made of is harder than wood
  - D. (d) because the heel pressure is too high ✓

**EXPLANATION**

D) Pressure ( $P = F/A$ ) is high due to small heel area (A).

- 13.** A point particle moves along a curvilinear path. Is it possible that its acceleration is zero?
- A. (a) Yes, but only if its vector speed is constant
  - B. (b) No, because there is always at least centripetal acceleration ✓
  - C. (c) Yes, but only if its scalar speed is constant
  - D. (d) No, because there is always at least tangential acceleration

**EXPLANATION**

B) Acceleration can't be zero in curvilinear motion (centripetal acceleration exists).

- 14.** Which of the following statements is true regarding the center of mass of a system of particles?
- A. The center of mass is always located at the geometric center of the system
  - B. The center of mass is the point where the total mass of the system can be considered to be concentrated ✓
  - C. The center of mass must be located within the physical boundaries of the system
  - D. The center of mass position depends on the choice of coordinate system
  - E. The center of mass velocity is always zero for a system with no external forces

**EXPLANATION**

B) Center of mass is the weighted average position of all mass in the system.

- 15.** In an infinitesimal adiabatic transformation the work
- A. is an exact differential
  - B. is an exact differential only if the system temperature is close to 0 K
  - C. is an exact differential only if it is a perfect gas
  - D. is never an exact differential ✓

**EXPLANATION**

Work in adiabatic transformations is path-dependent, hence not an exact differential.,

- 16.** Frictional forces (we think of the motion of an object in a fluid, air, water etc. ... for example) in general are not conservative. Can they be conservative in some particular cases?
- A. yes, if the motion is laminar and no vortices are formed ✓
  - B. Yes, if you neglect the form factor
  - C. no, in no case
  - D. yes, if the forces are strictly proportional to the speeds

**EXPLANATION**

Frictional forces can be conservative in laminar flow without vortices.,

17. Which of the following is not an intensive property?

- A. 1. Volume ✓
- B. 2. Temperature
- C. 3. Pressure
- D. 4. Velocity

**EXPLANATION**

Volume depends on system size, unlike intensive properties (temperature, pressure, velocity).,

18. The derivative of a unit vector:

- A. 1. is a vector orthogonal to the unit vector derivative. ✓
- B. 2. is a vector tangent to the trajectory.
- C. 3. is always zero in polar coordinates.
- D. 4. is an unit vector orthogonal to the unit vector derivative.

**EXPLANATION**

Derivative of a unit vector is orthogonal to the vector itself.

19. For a particles system on which no external forces act. It is always true that

- A. (A) the velocity vector of the center of mass of the system is necessarily zero.
- B. (B) the velocity vector of the center of mass of the system is constant. ✓
- C. (C) the vector momentum of the center of mass of the system is necessarily zero.
- D. (D) the vector angular momentum of the system with respect to the center of mass is necessarily zero.

**EXPLANATION**

Center of mass velocity is constant if no external forces act.

20. When a force  $F$  is conservative?

- A. 1. When  $\nabla \times F = 0$  ✓
- B. 2. When the integral on a generic path is zero.
- C. 3. When the work done by the force is zero.
- D. 4. When the force doesn't depend from time.

**EXPLANATION**

A force is conservative if its curl ( $\nabla \times F$ ) is zero.

21. In the absence of friction, would a pendulum continue to oscillate indefinitely?

- A. (a) No, because the motion of the pendulum is not a uniform motion
- B. (b) Yes, because the pendulum energy cannot dissipate. ✓
- C. (c) Yes, because the period of oscillation of the pendulum is constant
- D. (d) No, because the weight force is non-conservative

**EXPLANATION**

Without friction, energy is conserved, so pendulum oscillates indefinitely.,

22. Which of the following is not an intensive property? 1. Volume 2. Temperature

- A. Volume ✓
- B. Temperature

**EXPLANATION**

Volume depends on system size, unlike intensive properties (temperature).,

**23.** Which of the following is correct for a simple pendulum?

- A. Potential is minimum at the highest point.
- B. Kinetic energy is maximum at the highest point.
- C. Potential energy is minimum at the lowest point. ✓
- D. Total energy is minimum at the lowest point

**EXPLANATION**

Potential energy is minimum at the lowest point (equilibrium position).,

**24.** By introducing in a calorimeter at room temperature 300g of water at 353K it is observed that at equilibrium the temperature is 323K. Is it possible to deduce from this the thermal capacity of the calorimeter?

- A. (a) No, you need to know the nature of the various parts that make up the calorimeter, their specific heats, etc.
- B. (b) Yes ✓
- C. (c) No, is not an exact differential
- D. (d) No, er data are required

**EXPLANATION**

**Key fact:** Calorimeter heat capacity can be deduced from temperature changes.,

**25.** An aluminum block of mass  $m_1 = 0.1$  Kg at temperature  $T_1 = 580$  K is immersed in a glass vessel of mass  $m_2 = 0.2$  Kg and having temperature equal to  $T_2 = 300$  K. The glass vessel containing a mass of water equal to  $m_3 = 0.5$  Kg to the temperature of 300 K. Disregarding the exchange of heat with the outside environment, determine the equilibrium temperature  $T_F$  of the system. The specific heats of aluminum, glass and water are  $c_1 = 896.0$  J/Kg K,  $c_2 = 630.4$  J/Kg K and  $c_3 = 4187.0$  J/Kg K.

- A. (a)  $T_F = 410.86$  K
- B. (b)  $T_F = 310.86$  K ✓
- C. (c)  $T_F = 390.86$  K
- D. (d)  $T_F = 440.00$  K

**EXPLANATION**

**Key step:**  $m_1 c_1 (T_1 - T_F) = m_2 c_2 (T_F - T_2) + m_3 c_3 (T_F - T_2) \rightarrow T_F = 310.86$  K.

**26.** Determine the final temperature  $t$  of a mixture, when a mass  $m_1 = 100$  g of ice (fusion heat  $c_i = 80$  cal/g) at temperature  $t_1 = 0$  °C is mixed with a mass  $m_2 = 500$  g of water at temperature  $t_2 = 50$  °C, by assuming absent the dispersion of heat with the external environment ( $c_w = 1$  cal/g is the specific heat of water).

- A.  $t = -28.33$  °C
- B.  $t = -18.33$  °C
- C.  $t = -25.00$  °C
- D.  $t = 3.33$  °C ✓

**EXPLANATION**

**Key step:**  $m_1 L_f + m_1 c_w (T_F - 0) + m_2 c_w (T_F - 50) = 0 \rightarrow T_F = 3.33$  °C.

**27.** A mass  $M = 0.25$  Kg of copper at a temperature  $T_i$  is immersed in a container containing 0.1 kg of water initially at a temperature of 320 K. When the system reaches thermal equilibrium, 0.09 kg of water remains in the container. Determine the initial copper temperature  $T_i$ , ignoring the heat exchange with the external environment. Let  $c_w = 387$  J/Kg K and  $c_{Cu} = 4187$  J/Kg K be the specific heats respectively of copper and water. The latent heat of boiling water is worth  $\lambda_{(wb)} = 22.6 \cdot 10^5$  J/Kg.

- A.  $T_i = 640.00$  K
- B.  $T_i = 541.11$  K ✓
- C.  $T_i = 841.11$  K
- D.  $T_i = 441.11$  K

**EXPLANATION**

**Key step:**  $m_{Cu} c_{Cu} (T_i - T_F) = m_w c_w (T_F - T_w) + \lambda_{(wb)} (m_w - m_{w'}) \rightarrow T_i = 541.11$  K.

**28.** A homogeneous solid cylinder of length  $l = 40$  cm, radius  $r = 2$  cm and a mass  $m = 2$  Kg rotates in a horizontal plane around a vertical axis perpendicular to its axis in the middle point O as a result of an impulse  $J = 5 \times 10^{-2}$  Ns received perpendicular to an extreme. As a result of the friction the cylinder stops after traveling a lap. The initial angular speed  $\omega_0$  of the cylinder is  $3.7$  s $^{-1}$ . Calculate the moment  $M$  of the friction forces.

- A.  $M = 2.93 \times 10^2$  Nm
- B.  $M = 2.93 \times 10^4$  Nm
- C.  $M = 2.93$  Nm
- D.  $M = 2.93 \times 10^{-3}$  Nm ✓

**EXPLANATION**

**Key step:**  $M = J\omega_0 / \theta = 2.93 \times 10^{-3}$  Nm (using angular deceleration).

**29.** A pencil of length  $L = 15$  cm and mass  $m$  is placed in a vertical position on a plane with friction. Initially stopped, it falls by rotating around the point of contact with the plane. Calculate angular speed  $\omega$  at the impact instant with the plane.

- A. I cannot answer because I don't know the pencil mass.
- B.  $\omega = 0.14$  s $^{-1}$
- C.  $\omega = 14$  s $^{-1}$
- D.  $\omega = 1.40$  s $^{-1}$  ✓

**EXPLANATION**

**Key step:**  $\omega = \sqrt{(3g/L)} \approx 1.40$  s $^{-1}$  (using energy conservation).

**30.** Archimedes' thrust on a body partially immersed in a liquid is applied

- A. at the center of gravity of the supposedly homogeneous body
- B. at the center of gravity of the body
- C. at the center of gravity of the mass of fluid moved ✓
- D. there is no point of application in the case of pressure

**EXPLANATION**

Archimedes' thrust acts at the center of gravity of the displaced fluid.,

**31.** The amount of heat absorbed by a system during an infinitesimal transformation is not in general an exact differential. Can it be in some particular cases?

- A. No, in no case
- B. Yes, if the transformation is reversible ✓
- C. Yes, if it is a perfect gas
- D. Yes, if the work done by the system is an exact differential

**EXPLANATION**

Heat is an exact differential only for reversible transformations.,

**32.** To calculate the orbit of a planet around the sun in addition to the mass of the planet and the sun, it is necessary and sufficient to know:

- A. the total energy of the system
- B. the total energy and the total angular momentum of the system ✓
- C. the radius of the planet
- D. the total angular momentum of the system

**EXPLANATION**

**Key fact:** Total energy and angular momentum determine orbital motion.,

**33.** Which of the following statements is correct? (a) The entropy of a hot copper block decreases as it cools. (b) If ice is melted in water in an insulated container, the net entropy decreases. (c) The entropy of an isolated system must remain constant or increase. (d) Work must be input if energy is transferred from a cold body to a hot body.

- A. The entropy of a hot copper block decreases as it cools.
- B. If ice is melted in water in an insulated container, the net entropy decreases.
- C. The entropy of an isolated system must remain constant or increase. ✓
- D. Work must be input if energy is transferred from a cold body to a hot body.

**EXPLANATION**

Entropy of an isolated system never decreases (2nd Law of Thermodynamics).,

**34.** A perfect gas thermal machine describes a cycle formed by a reversible isothermal expansion, a reversible isochoric transformation, an irreversible isothermal compression and a reversible isochoric transformation. In a cycle

- A. the entropy variation of the thermodynamic universe is negative
- B. the entropy variation of the thermodynamic universe is zero ✓
- C. the entropy variation of the gas is positive
- D. the entropy variation of the gas is zero

**EXPLANATION**

Entropy change of the universe is zero for a reversible cycle.,

**35.** How is changing the pressure  $p$  in the atmosphere with the altitude  $z$ , if the atmosphere can be considered as a perfect gas? (Here  $a$  is constant)

- A.  $p(z) = p(z = 0)gz$  where  $p$  is the air density
- B.  $p(z) = p(z = 0)(z/a)$
- C.  $p(z) = p(z = 0)\exp(-z/a)$  ✓
- D.  $p(z) = p(z = 0)\exp(z/a)$

**EXPLANATION**

**Key formula:**  $p(z) = p(0)\exp(-z/a)$  for hydrostatic equilibrium in a perfect gas.

**36.** Under the action of an unknown force field, a point mass of 3g moves frictionlessly along the curve of equation  $x(t) = 2t$ ,  $y(t) = t^2$ ,  $z(t) = t^3$  where  $t$  is the time. Is it possible to calculate the work done by the forces in any given generic time?

- A. (a) Yes, provided that the force field is conservative
- B. (b) Yes, in any case ✓
- C. (c) No, because the force field cannot admit potential
- D. (d) No, because the force field is not known

**EXPLANATION**

Work can be calculated via line integral if the path is known, regardless of field.

**37.** Consider  $n = 0.20$  moles of nitrogen (a diatomic gas) initially at temperature  $T_1 = 300$  K and pressure  $P_1 = 10^5$  Pa. The gas performs a reversible adiabatic compression until the pressure reaches  $P_{(F)} = 2P_1$ . Calculate the final volume  $V_{(F)}$ , the final temperature  $T_{(F)}$ , and the work  $W$  done by the gas.

- A.  $V_F = 3.0 \cdot 10^{-3} \text{ m}^3$ ;  $T_{(F)} = 300$  K;  $W = -254$  J
- B.  $V_F = 3.0 \cdot 10^{-3} \text{ m}^3$ ;  $T_{(F)} = 329$  K;  $W = -254$  J
- C.  $V_F = 3.0 \cdot 10^{-3} \text{ m}^3$ ;  $T_{(F)} = 361$  K;  $W = -254$  J ✓
- D.  $V_F = 3.0 \cdot 10^{-3} \text{ m}^3$ ;  $T_{(F)} = 361$  K;  $W = -154$  J

**EXPLANATION**

**Key steps:**  $V_{(F)} = V_1(P_1/(P_{(F)}))^{1/\gamma} = 3.0 \times 10^{-3} \text{ m}^3$ ;  $T_{(F)} = T_1((P_{(F)})/P_1)^{((\gamma-1)/\gamma)} = 361$  K;  $W = nC_v(T_{(F)} - T_1) = -254$  J.

**38.** A sphere of 10g hits centrally and elastically, at a speed of 5m/s another stationary sphere of unknown mass. After the impact, it goes back with a speed of  $2\frac{m}{s}$ . Is it possible, neglecting the frictions, to calculate the mass of the impacted sphere?

- A. (a) No, because in this type of impact the total energy of the system composed of the two spheres is not conserved
- B. (b) No, because the speed of the second sphere hit after the impact is unknown
- C. (c) Yes ✓
- D. (d) No, a collision without friction a nonsense

**EXPLANATION**

**Key step:** Conservation of momentum + kinetic energy  $\rightarrow m_2 = 0.071$  kg.

**39.** A simple pendulum used to measure time is made of a mass attached to the end of a steel wire. The pendulum correctly shows 1 s when the temperature is  $0^\circ\text{C}$ . Calculate how much time the clock will lose per day in a hot country where the average temperature is  $40^\circ\text{C}$ ? (The linear expansion coefficient of steel is equal to  $1.6 \times 10^{-5} \text{ }^\circ\text{C}^{-1}$ )

- A. The pendulum lags by 20 s per day
- B. The pendulum lags by 2.0 s per day ✓
- C. The pendulum lags by 10 s per day
- D. The pendulum lags by 1.0 s per day

**EXPLANATION**

**Key step:**  $\Delta t/\text{day} = (1/2)\alpha\Delta T \times 86400 \text{ s} \approx 2.0 \text{ s/day}$ .

**40.** The angular momentum of a system of particles is conserved

- A.** if the sum of the external torques acting on all the particles is zero. ✓
- B.** if the particle are all equal.
- C.** if the internal torque on each particle is zero.
- D.** if the sum of the internal torques is zero.

**EXPLANATION**

Angular momentum is conserved if net external torque is zero.,

**41.** Two bullets of different mass are fired with the same speed, with the same inclination and from the same point. Neglecting the air resistance and assuming the perfectly flat ground surface, which takes less time to get to the ground?

- A.** (a) the one with less mass
- B.** (b) the one with the greatest mass
- C.** (c) they will both take the same amount of time ✓
- D.** (d) the data are insufficient to reply

**EXPLANATION**

Time of flight depends only on vertical motion (same for both masses).,

**42.** The axis of a spinning top rotates around the vertical axis that passes through the point of contact with the ground. This fact is due

- A.** at the torque of gravity force relative to the point of the spinning top in contact with the ground. ✓
- B.** at the torque of gravity force relative at the center of mass of the spinning top.
- C.** at the moment of the frictional force that the ground exerts on the spinning top.
- D.** to the variation of the gravity force between the ends of the spinning top.

**EXPLANATION**

Precession is due to torque of gravity about the contact point.,

**43.** A steel ball is released from a height  $h$ , falls to the ground and bounces back. Neglecting the friction with air, the height it will attain after the rebound

- A.** 1. is equal to  $h$  only if the collision with the ground is perfectly elastic. ✓
- B.** 2. is equal to  $h$  if the momentum is conserved during the collision with the ground.
- C.** 3. is equal to  $h$ .
- D.** 4. is equal to  $h/2$ .

**EXPLANATION**

Rebound height equals initial height only for perfectly elastic collisions.,

## Answer Key

|       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|
| 1. C  | 2. A  | 3. C  | 4. A  | 5. A  | 6. A  |
| 7. D  | 8. C  | 9. D  | 10. B | 11. A | 12. D |
| 13. B | 14. B | 15. D | 16. A | 17. A | 18. A |
| 19. B | 20. A | 21. B | 22. A | 23. C | 24. B |

|              |              |              |              |              |              |
|--------------|--------------|--------------|--------------|--------------|--------------|
| <b>25. B</b> | <b>26. D</b> | <b>27. B</b> | <b>28. D</b> | <b>29. D</b> | <b>30. C</b> |
| <b>31. B</b> | <b>32. B</b> | <b>33. C</b> | <b>34. B</b> | <b>35. C</b> | <b>36. B</b> |
| <b>37. C</b> | <b>38. C</b> | <b>39. B</b> | <b>40. A</b> | <b>41. C</b> | <b>42. A</b> |
| <b>43. A</b> |              |              |              |              |              |